HIKET — the storyline

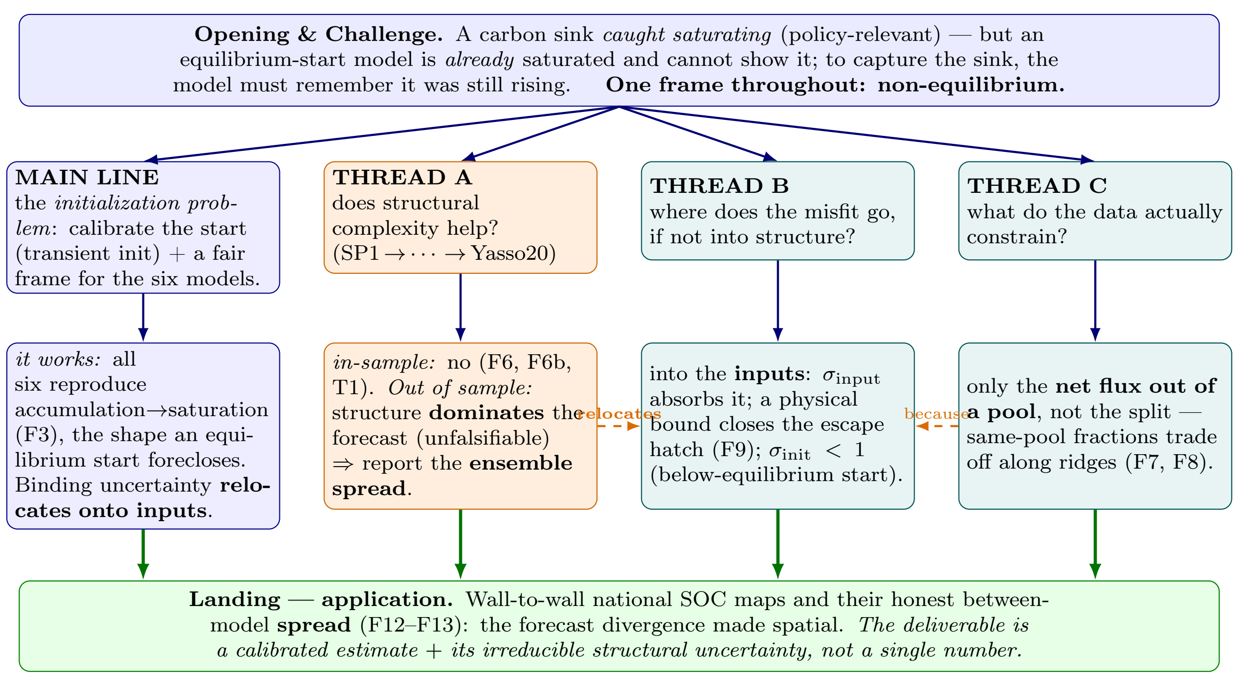
a working note for coauthors

Version of August 13, 2026 — a narrative testbed, not the manuscript. No methods or full referencing; the point is to read the argument whole and see how it flows.

DRAFT STATUS — 2026-08-10. All figures and numbers in this note are keyed to the `20260807_1655*` posteriors and are **provisional**. A recalibration is in flight (Roihu jobs 563524–563529, commit `3b0d533`) which corrects the likelihood error scale, the Fortran precision of Yasso15/20, and three prior widths misread from their sources. Two findings that will need incorporating either way: **TP3's non-convergence is a result, not a defect** — its two modes fit within 4.2 nats, so the third pool is not identifiable from two SOC observations per plot; and **bulk residence times must be quoted from the intrinsic basis** (`doublechecks/intrinsic_mrt.R`), not from the earlier contaminated computation.

What this is. Below is the HIKET argument told as one continuous story, with each figure and table placed where the story needs it. It follows an Opening–Challenge–Action–Resolution arc with three side threads; the schema on the next page is the map. The prose is deliberately compact — enough to feel the flow and judge whether it lands — and is the scaffold the manuscript will be written onto.

The story at a glance



Two threads run throughout: inference *(the unknown start becomes a calibrated parameter whose uncertainty propagates) and* history *(Finnish forest management explains the non-equilibrium and the rising inputs).*

The through-line is one physical fact — **these soils are not at equilibrium** — seen from two sides: it is why there is a sink to report (the stakes) and why a conventional model cannot show it (the obstacle). The whole paper is the work of turning the second into a solution for the first.

1 Opening: a sink caught saturating, and a model that cannot see it

Finnish forest soils have been accumulating carbon through the observation period and are now bending toward a plateau. This incipient saturation is not a modelling curiosity: Finland reports it in its national greenhouse-gas inventory, where most countries assume a flat soil pool, so getting it right has direct policy weight. The signal has a physical cause. Across the twentieth century the country's forests recovered from a depleted, heavily-used early-century base, and the growing stock has risen by roughly seventy percent since the 1970s (Fig. 1); the litter those forests return to the soil rose with them and then levelled off, and it varies strongly from south to north (Fig. 2). A rising, non-stationary input history feeding soils that were well below their carrying capacity is exactly the setup for an accumulating, saturating sink.

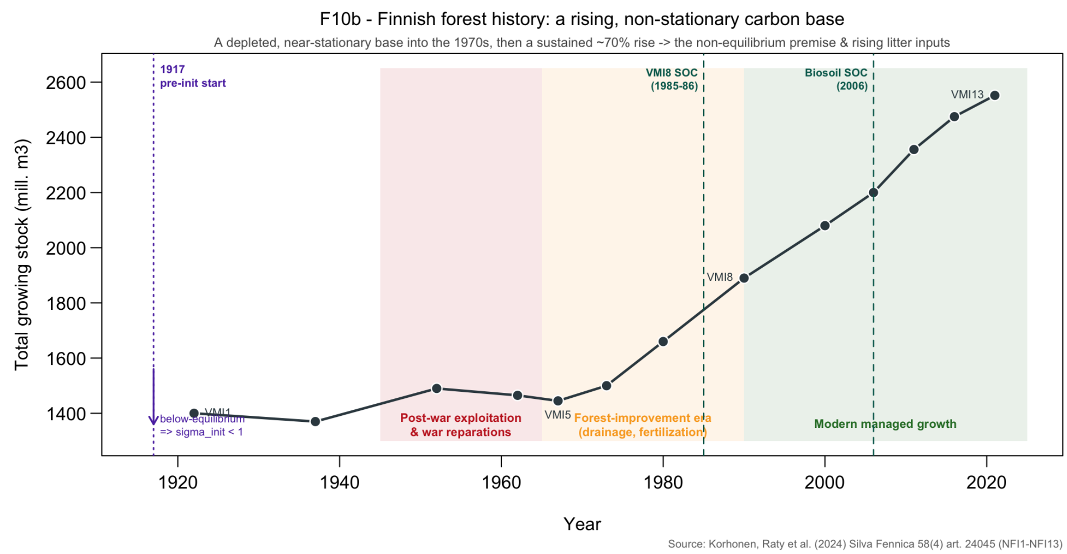


Figure 1: The non-equilibrium premise. National growing stock (NFI, 1921–2023): a depleted, near-stationary base into the 1970s, then a sustained recovery. The 1917 pre-initialisation anchor and the two SOC campaigns are marked. Source: Korhonen, Räty et al. (2024).

The trap is that a soil model run the conventional way cannot represent any of this. Started at steady state — the near-universal default — the model already sits at its saturated end-state, with nowhere left to accumulate. Run forward at published parameters from an equilibrium start, it stays flat, and well below the observed stocks (Fig. 3). The very phenomenon we most want to capture is the one the standard convention forecloses.

2 Challenge: to show the saturation, the model must remember it was still rising

The stakes and the obstacle are the same fact from two sides. The soils hold an accumulating sink *because* they are out of equilibrium; the standard model fails *because* it assumes they are not. So capturing the observed dynamic is not a matter of better parameters — it requires abandoning the equilibrium-start convention and initialising the model as a system still recovering from its history. That is the initialization problem, and it is the axis the whole study turns on: how do you start a non-equilibrium simulation when you have no record of the forcing that produced the initial state?

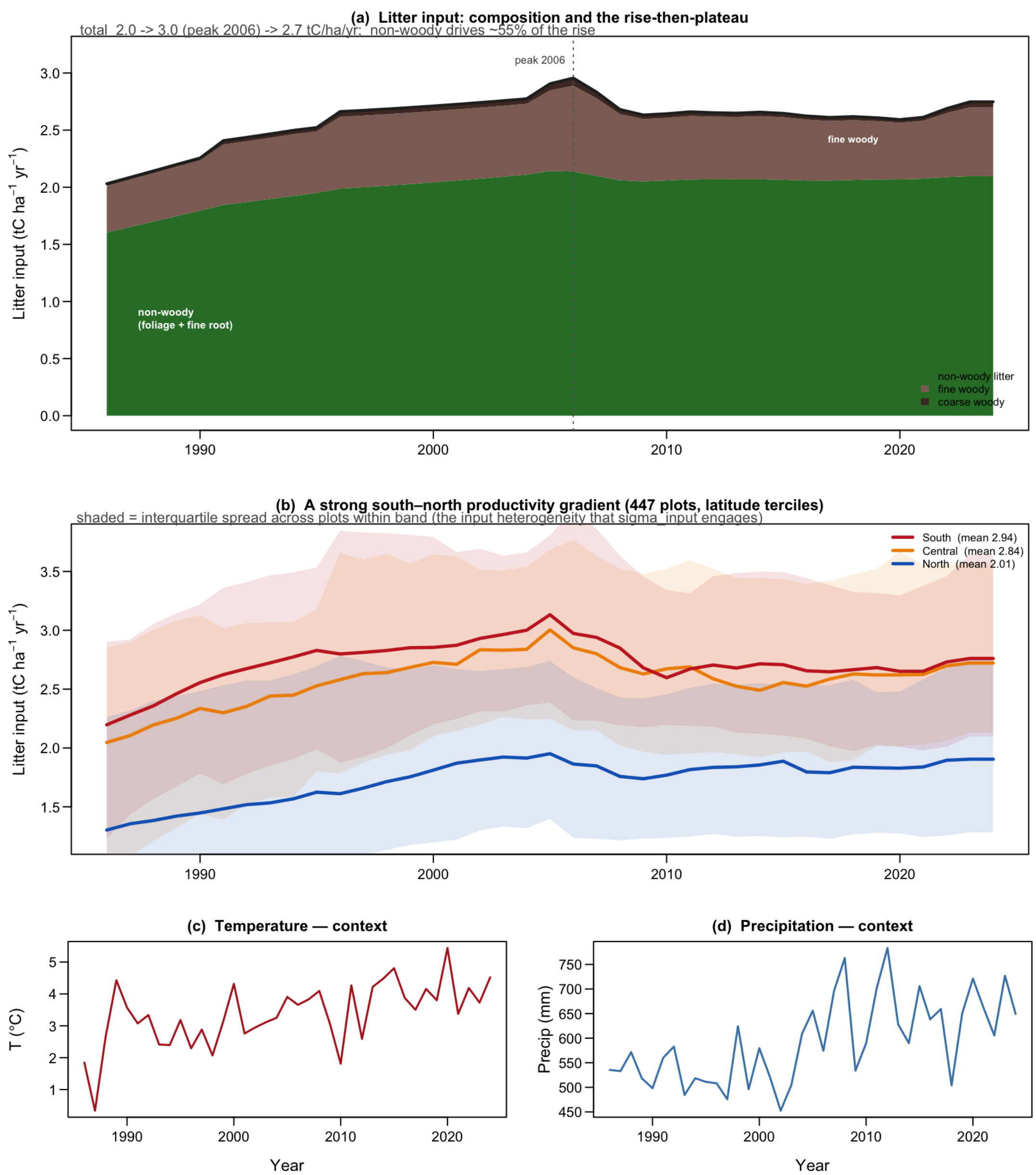


Figure 2: The inputs. Litter forcing over time (composition; a rise to a mid-2000s peak, then a plateau) and its strong south–north gradient, with climate shown only as context. The accumulation is input- and history-driven, and the inputs vary enormously in space.

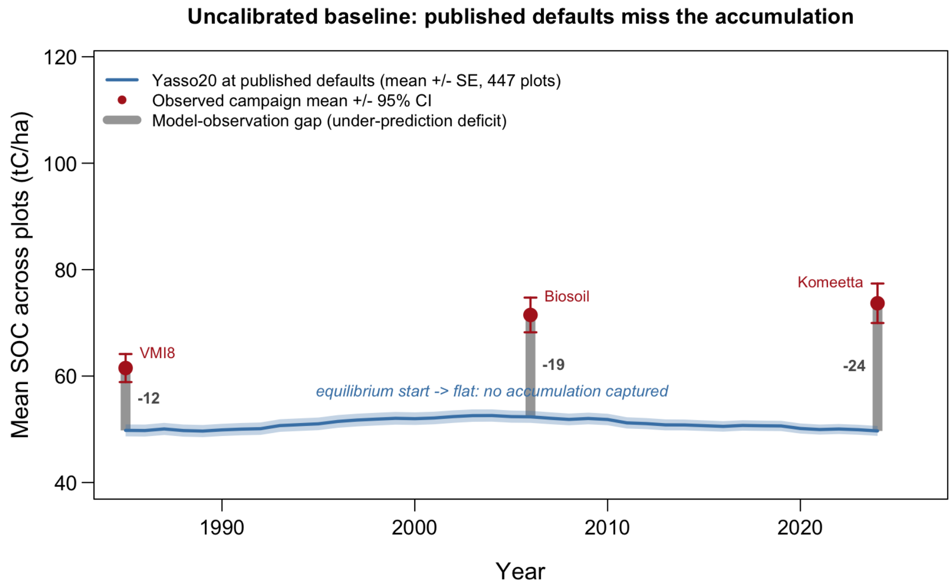


Figure 3: The trap. Yasso20 at published defaults with an equilibrium start, averaged over all plots: flat, because an equilibrium start cannot accumulate, and low relative to the observed campaign means.

3 Action: transient initialization under a fair frame

The move is to stop *assuming* the initial state and instead *calibrate* it. Each model is spun up from a below-equilibrium historical anchor to the start of the observation period, and the initial carbon becomes a calibrated quantity, σ_{init}, whose uncertainty propagates into both the starting state and the forward run rather than being fixed by fiat. Casting the unknown start as an inferred parameter, not a fixed assumption, is what makes this tractable at scale.

The comparison across models only means something if the models are not secretly competing on who guessed the inputs or the initial state best. We therefore put all six on an even footing: homogenised priors (identical in construction, not in raw number), equal degrees of freedom in the inter-pool flow fractions, and two auxiliary parameters — σ_{init} for the initial state and σ_{input} for the litter level — that absorb init- and input-uncertainty identically for every model. The six span a deliberate complexity ladder: a one-pool model (SP1), two- and three-pool cascades (TP2, TP3), and the three five-pool Yasso variants, with a second, climate-integration axis running Yasso07 $\rightarrow$ 15 $\rightarrow$ 20. On this even ground, structure — and only structure — is what varies.

4 Resolution: it works, and the binding uncertainty relocates

4.1 The payoff

Transiently initialised, all six models reproduce the observed shape — accumulation bending toward incipient saturation — and track the later campaign means (Fig. 4). This is the shape an equilibrium start cannot produce: solving the initialization problem is precisely what lets the models represent the policy-relevant sink. The payoff comes with an honest caveat visible in the reconstructed trajectory (Fig. 5): the models converge on a 1985 stock some 26 tC ha^{-1} above the observed start, a consequence of the coarse, unrecorded pre-1985 phase — a live question, not a failure of the method.

Reading note — the SOC observations have been rebuilt; these figures predate the rebuild. The campaign points in this and the next figure are *whole-profile* carbon — organic layer plus mineral soil, not the 0–40 cm inventory total — but they were produced from our earlier SOC assembly, which we have since replaced. Two defects were found (2026-08-03) and corrected. First, the mineral layers carried no coarse-fragment (stoniness) correction, which inflated mineral carbon by a near-constant factor of ≈1.6; with a median stoniness of 43%, $1/(1-0.43)$ accounts for it. Second, the three campaigns were processed along separate paths that had drifted apart, so part of the 1985→2006 change was protocol rather than dynamics.

All three campaigns now come from one LUKE source (organic + mineral, stoniness- and bulk-density-corrected), reproducing the official national stocks exactly, with the deep tail extrapolated per plot only as far as augering actually reached rather than to a flat 1 m. The corrected target rises gently — campaign medians 59.4, 66.7, 67.4 tC ha^{-1} — where the old one gave 63, 102, 105, i.e. a 62% jump between 1985 and 2006 that was an artefact. Section **??** of the main manuscript documents this in full.

What this means for the figures below. They stand as the argument's *shape* — the accumulation-then-saturation, the model ordering, the initialization gap — but their SOC levels, and any number quoted against a campaign mean (including the 26 tC ha^{-1} 1985 offset above), are tied to the superseded target and will be restated once the six models are recalibrated on the corrected data. Read the levels as provisional; read the structure as the result.

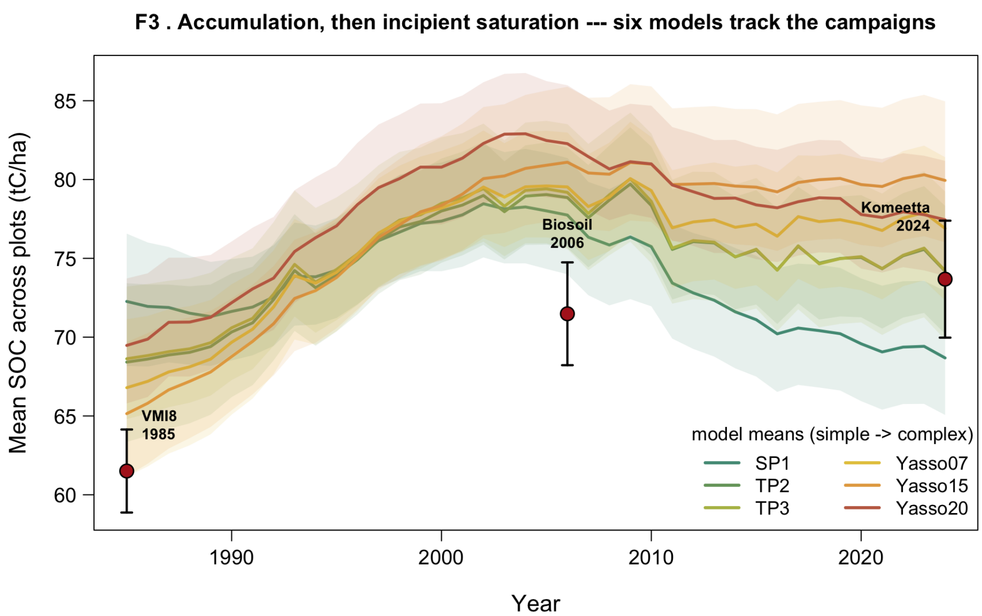


Figure 4: The central result. Cross-plot mean SOC, all six models, 1985–2024: rise then bend toward saturation, tracking the Biosoil/Komeetta campaign means. Colour runs from the simple models (green) to the Yasso family (warm).

Reading note — depth basis. As in Fig. 4, the observed campaign stocks (panel a) are whole-profile values (measured organic + 0–40 cm, plus a modelled deep tail), and they too come from the superseded SOC assembly described in the note above. Levels provisional, structure robust.

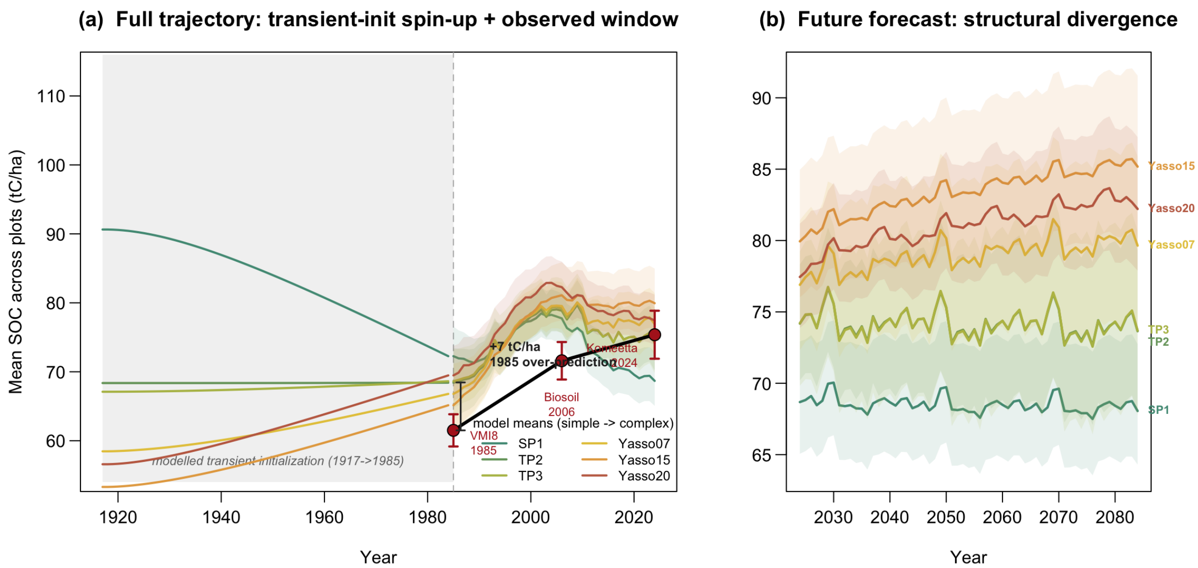


Figure 5: The initialization problem, made visible. (a) The full trajectory including the reconstructed transient-init spin-up 1917→1985 and the observed window; the models recover from a below-equilibrium start but disagree about how much, and all reach $\sim$26 tC ha^{-1} above the observed 1985 stock. (b) The future forecast 2025→2084: agreement where calibrated, sharp divergence when extrapolated.

4.2 Thread A — does adding structural complexity improve the answer?

In-sample, no. Predictive skill does not climb the pool ladder or the climate sub-axis; the most complex model fits worst, on both calibration and independent holdout (Fig. 6, Fig. 7). We do make the anti-complexity claim, but scope-bounded: for an applied, recalibratable, single-region task whose aim is accuracy plus calibrated uncertainty, added structure does not buy skill and can cost it. That is not a verdict on Yasso, which is optimised for a different objective — generality, standardisation, and litter-quality resolution that this single-region fit test simply does not measure — so "more complexity does not help *here*" and "Yasso is the right inventory tool" coexist. Table 1 is the whole comparison at a glance.

Table 1: Results at a glance. Per-model structural complexity (pools, free parameters), skill (R^2 calibration and holdout, holdout RMSE), the input story (effective litter flux $\sigma_{\text{input}}\bar{J}$, all inside the physical $[0.5, 8.7]$ tC ha^{-1} yr^{-1} envelope), and the initial state ($\sigma_{\text{init}} < 1$). Complexity climbs while skill does not.

Model	Pools	Free params	R^2 (cal)	R^2 (hold)	RMSE (hold)	Eff. flux $\sigma_{\text{inp}}\bar{J}$	σ_{init}
SP1	1	6	0.007	0.002	53	4.5	1.11
TP2	2	7	0.005	0.002	55	5.9	0.81
TP3	3	9	0.005	0.001	55	5.9	0.78
Yasso07	5	20	0.012	0.006	54	6.2	0.64
Yasso15	5	26	0.019	0.017	54	5.0	0.53
Yasso20	5	26	0.013	0.009	55	6.0	0.53

The picture inverts out of sample. Extrapolated to 2084, the models agree where the data constrain them and then fan out sharply — the one-pool SP1 climbs without bound toward $\sim$133 tC ha^{-1}, while the stiffer Yasso structures plateau near 108 (Fig. 5b). The very structural choices the data cannot pin down now govern the projection, and in-sample skill is blind to this because it is scored only on

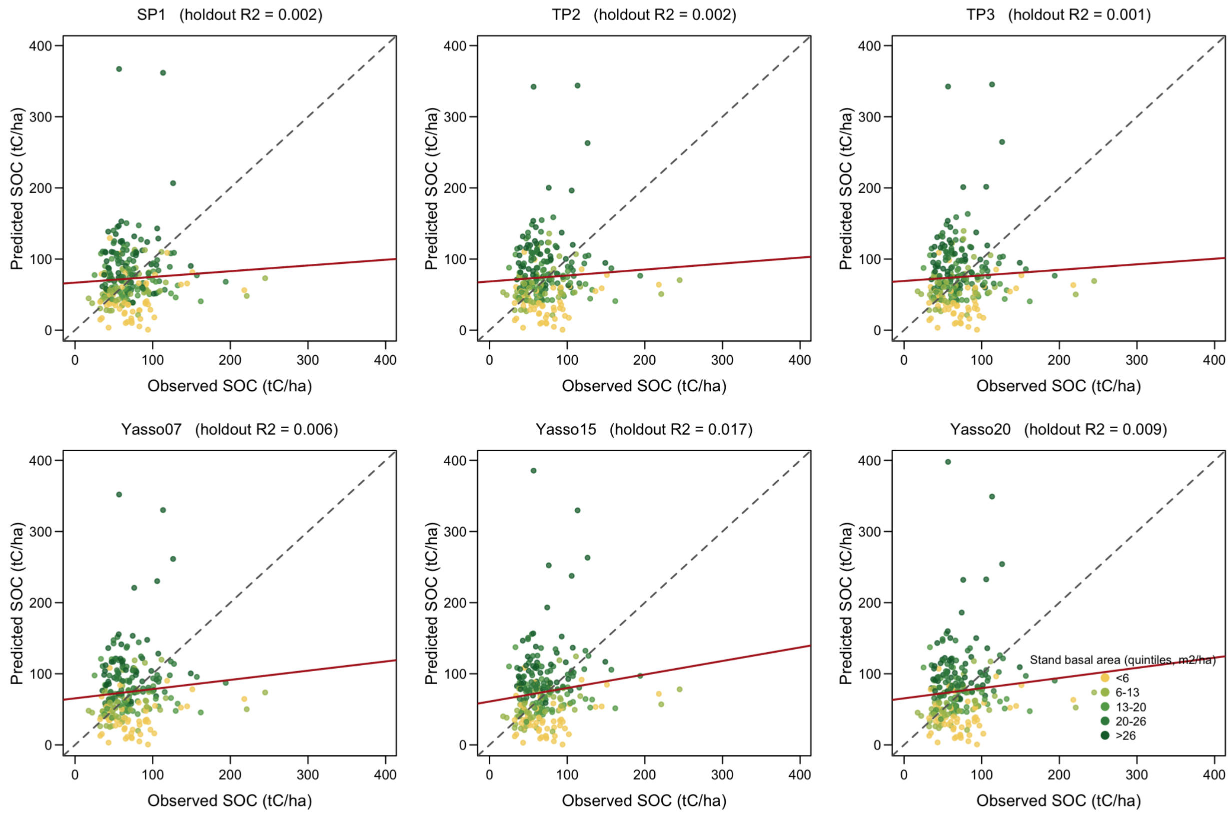


Figure 6: Skill on the honest set. Observed vs. predicted SOC on the held-out plots, by model, with points classed by stand basal-area quintile. Skill is uniformly low and the ranking nearly collapses out of sample; the red least-squares line is far flatter than the 1:1 dashed line (the models compress the range).

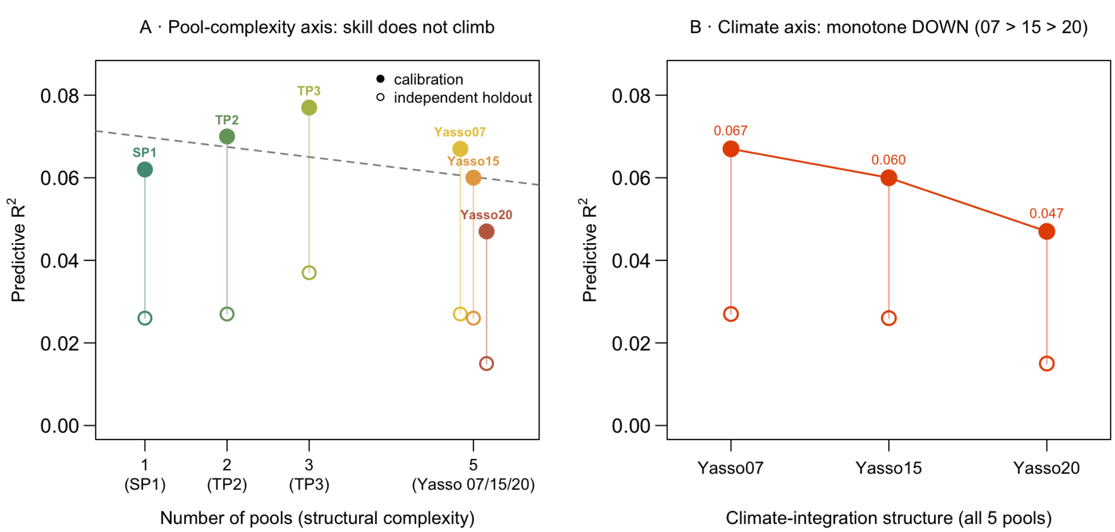


Figure 7: Complexity does not move the answer. Predictive R^2 against structural complexity: the pool-axis slope is negative and the climate sub-axis falls monotonically; the most complex model is the worst on both calibration and holdout.

the present stocks. Process plausibility gives a soft steer — soils saturate, so SP1's unbounded rise is the least credible trajectory — but it is soft, and the data cannot adjudicate. The decision-relevant quantity, multi-decade SOC change, is precisely the one in-sample skill cannot validate; the honest output is therefore the ensemble spread, not a single model.

4.3 Thread B — the binding uncertainty is the inputs

Where does the misfit go, if not into structure? Into the inputs. The multiplier σ_{input} is where each model absorbs what its kinetics cannot. Left as a weak, scale-free nuisance, the likelihood drags two models (TP2, TP3) to effective litter fluxes of 13–20× boreal net primary production — physically impossible. Bounding the effective flux to a physical envelope closes that escape hatch, and the diagnostic is decisive: under the bound all six fall inside the envelope, and *only* the two runaway models lose predictive skill (Fig. 8). That the skill drop is *selective* — bounding a shared input nuisance singles out exactly the two models that had been manufacturing carbon — is the cleanest evidence that inputs, not kinetics, were the lever. The companion parameter σ_{init} is uniformly below one: a below-equilibrium start, the forest history of Fig. 1 made quantitative.

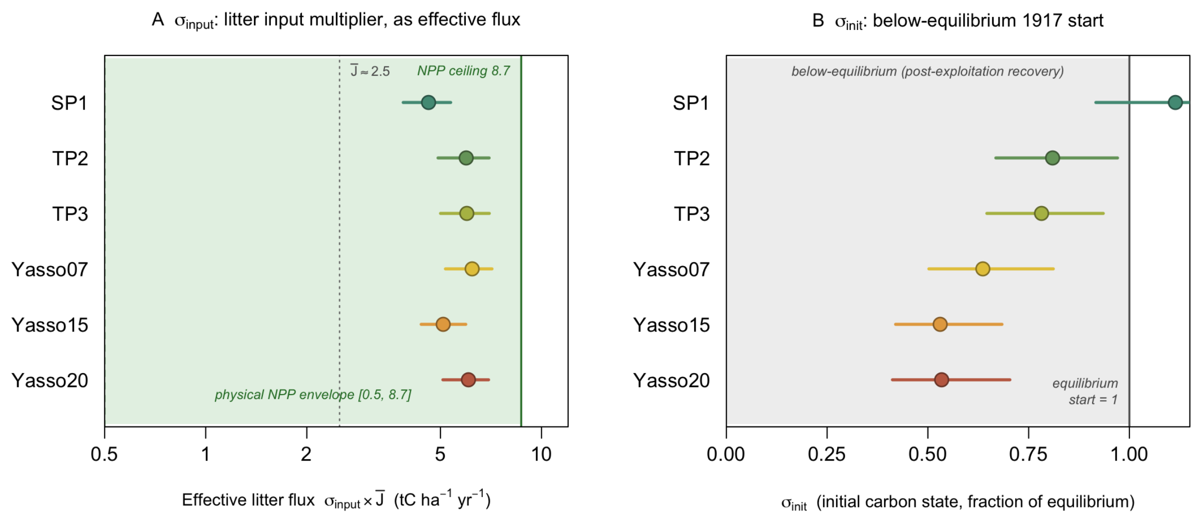


Figure 8: The auxiliary parameters, both physically interpretable. (a) σ_{input} as effective litter flux against the boreal NPP envelope (green); under the physical bound all six sit inside it. (b) σ_{init} as the initial carbon state; every interval lies below one, a post-exploitation, still-recovering start.

4.4 Thread C — what the data can and cannot constrain

The reason structure is a weak lever is that the data barely see it. The twelve inter-pool transfer fractions gain essentially no information from calibration — their posteriors stay on their priors (Fig. 9) — while σ_{input} gains the most. The mechanism is geometric (Fig. 10): fractions leaving the same pool anti-correlate almost perfectly, so the data constrain only the *net* flux out of each pool, never how it splits among destinations. Each source pool contributes one identified quantity; everything else is a ridge the posterior slides along (Table 2). This is the mechanism behind "complexity buys shape, not skill": the parameters that complexity adds are the ones the data cannot resolve. We do not treat this as a defect to be fixed — it is intrinsic to the model class and the two stocks per plot — but manage it openly, pinning rates where authoritative values exist (Yasso) and calibrating them where they do not (the simple models).

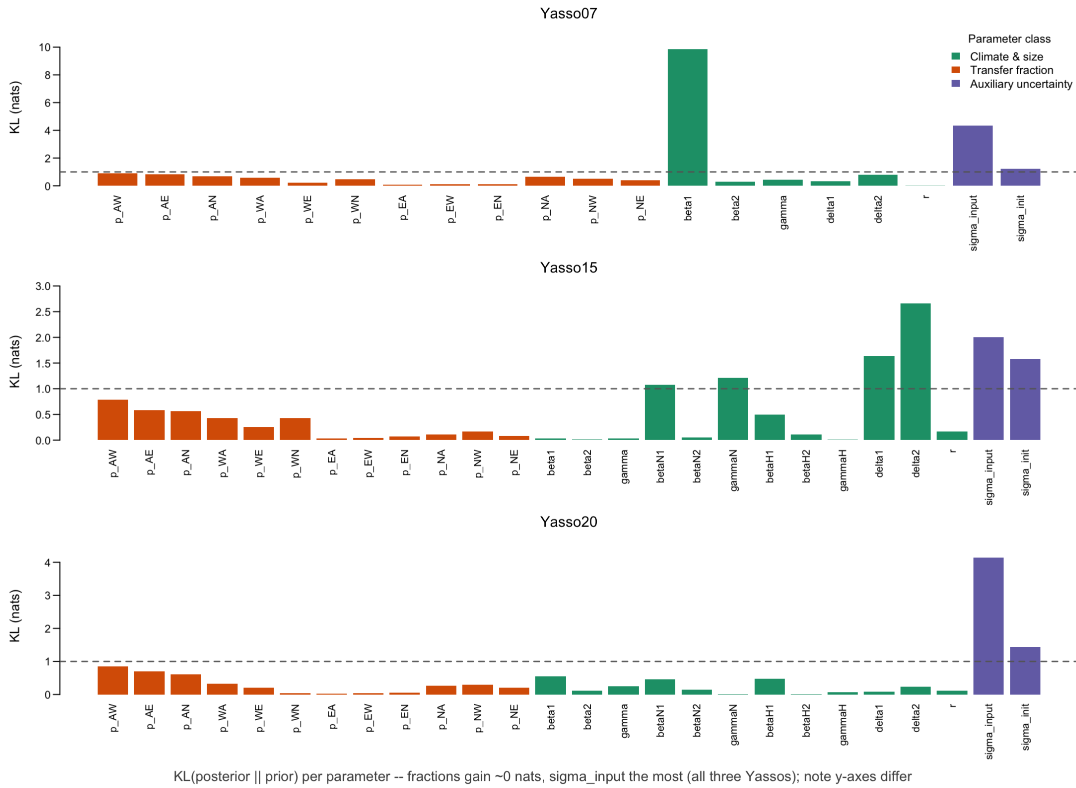


Figure 9: Information gain per parameter, the three Yasso models. The transfer fractions (orange) gain ≈0 nats; σ_{input} (purple) gains the most. The verdict — data constrain inputs, not kinetics — is the same across all three variants.

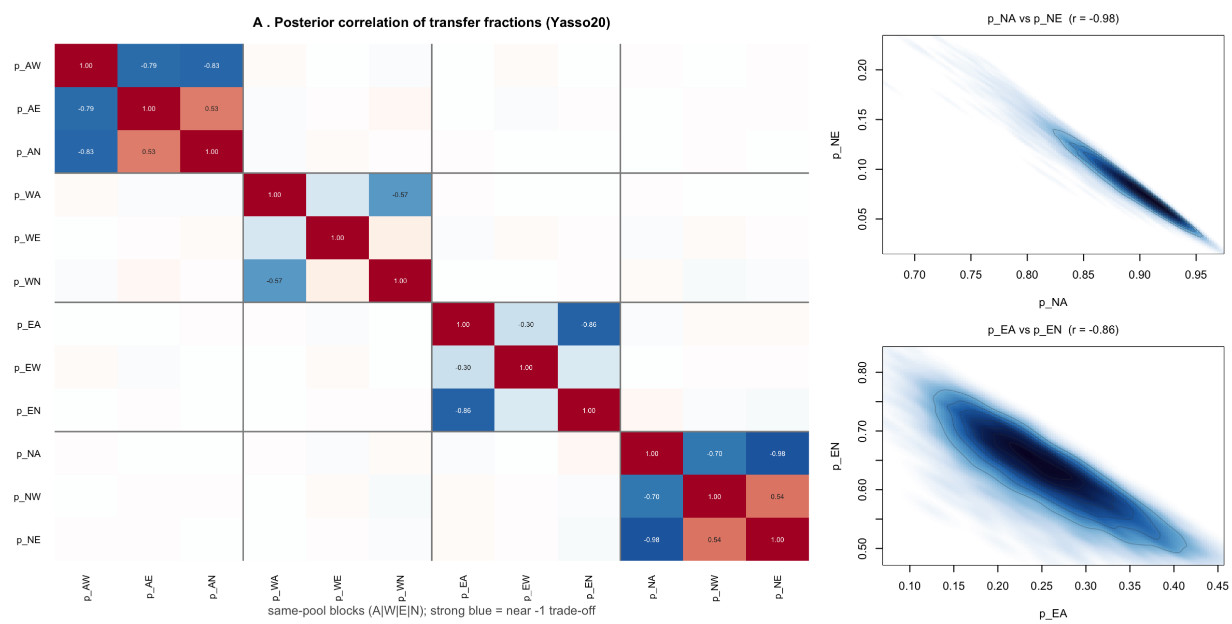


Figure 10: The trade-off structure (Yasso20). (a) Posterior correlations of the transfer fractions, blocked by source pool: same-pool fractions anti-correlate near -1; cross-pool correlations vanish. (b) Two same-pool ridges as continuous density fields. The data fix the net flux out of a pool, not the split.

Table 2: The strongest same-pool anti-correlations (Yasso20): each is a net-flux trade-off along which the posterior is unconstrained.

Source pool	Fraction pair	Posterior r	Interpretation
N	$p_{NA} \leftrightarrow p_{NE}$	-0.98	net-flux trade-off
E	$p_{EA} \leftrightarrow p_{EN}$	-0.86	net-flux trade-off
A	$p_{AW} \leftrightarrow p_{AN}$	-0.83	net-flux trade-off
A	$p_{AW} \leftrightarrow p_{AE}$	-0.79	net-flux trade-off
N	$p_{NA} \leftrightarrow p_{NW}$	-0.70	net-flux trade-off
W	$p_{WA} \leftrightarrow p_{WN}$	-0.57	net-flux trade-off

4.5 What a longer residence time would cost

The same geometry governs the quantity the reader will care about most: our mean residence times come out well below the published parameterisations. Fig. 11 shows why, and how much of it is the data's doing. In the Yasso family MRT and σ_{input} trade off along a clear ridge ($r = -0.79/-0.73/-0.57$): the observed stock pins their *product*, so the split between "slow turnover, modest inputs" and "fast turnover, inflated inputs" has to be decided by something other than the stock. Notably, the *simple* models show no such ridge ($r \approx -0.26/-0.31$), so this is a property of the Yasso structure, not of the calibration in general.

The cost of moving to the published residence time is then wildly uneven — so much so that **"our MRT is too short" is not one finding but three different ones**. Yasso20's posterior straddles its published value (18 % of draws exceed it) and sitting there costs 1.6 log-likelihood units, which is nothing. Yasso15 reaches its published value only in the far tail (0.011 % of draws) at a cost of 14.7. **Yasso07 never reaches its published value at all:** 225 015 draws span 10.4–25.4 yr against a published 33.5, so the posterior stops eight years short and the figure can put no price on the gap — only a profile likelihood could.

Which published value. Those costs are measured against the published *point* parameterisation (33.5 / 30.4 / 19.0 yr). Yasso15 and Yasso20 also publish posterior *samples*, and because residence time is a nonlinear many-to-one function of the parameters, the median residence time of that sample is not the residence time of the median parameters — for Yasso20 it is 25.0 rather than 19.0 yr, which raises its cost from 1.6 to 10.0. The qualitative ordering is unchanged, but the two comparators must not be mixed, and the figure uses the point throughout.

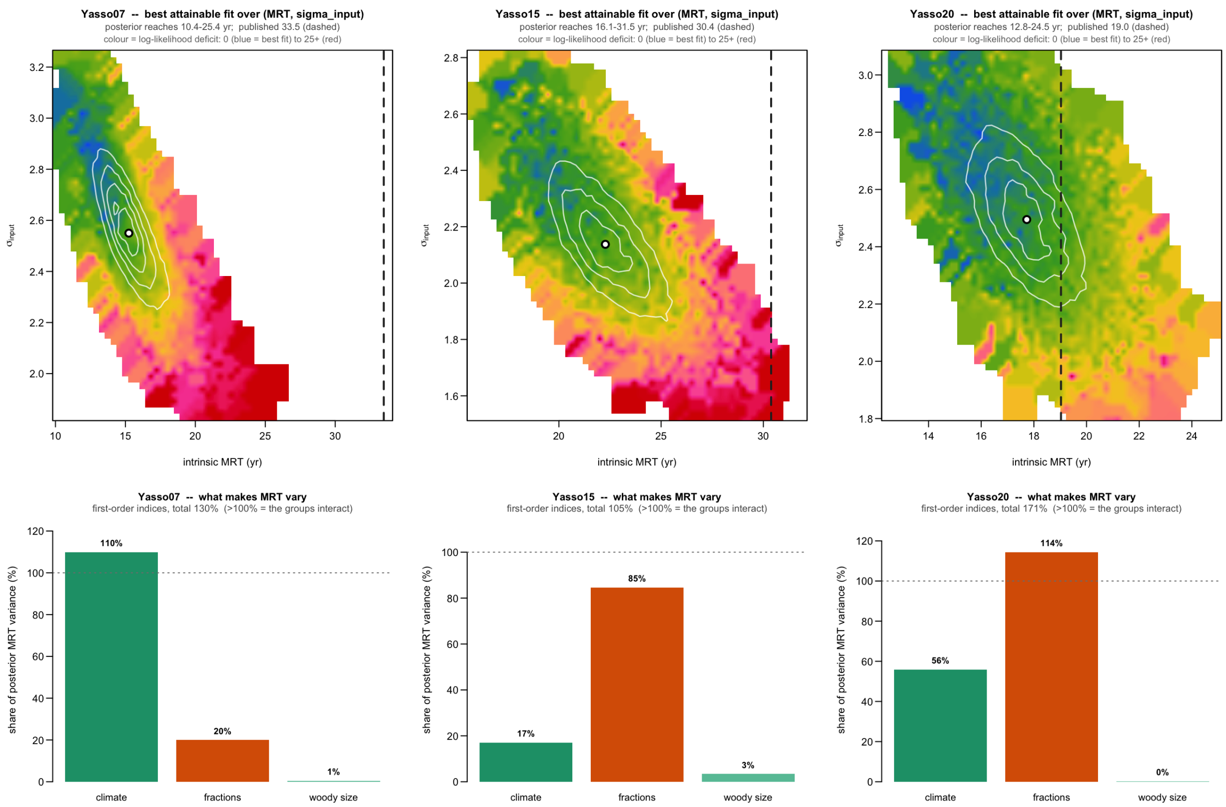


Figure 11: What a longer residence time would cost. *Top:* best attainable fit over (intrinsic MRT, σ_{input}) for each Yasso model. **Colours are the likelihood alone — the fit to the data, priors excluded — blue is the best fit, through green and yellow to red at the worst. The white contour lines are the posterior — likelihood *and* priors together.** Colour is the log-likelihood deficit from the best fit found, clamped at 25 units. Dashed line: published MRT; white dot: posterior median. Regions the sampler visited fewer than five times are left blank. *Bottom:* first-order variance indices of MRT. The colour peak lies up and to the left of the contours in all three panels: the data alone want an *even shorter* residence time, and it is the prior that pulls the answer back toward the published value.

Reading the two layers. The panels superimpose two different things on the same axes, and keeping them apart is the whole point of the figure.
The white contour lines are the posterior — where the calibration actually ended up. That location is set by three ingredients at once: how well the data are fitted, what the priors say, and how many parameter combinations happen to land there (many parameters map onto the same residence time, and that alone attracts probability mass).
The colours are the likelihood only — the best fit to the data achievable in that cell, with the priors switched off. They answer a different question: not "where did we end up", but "how well *could* the data be fitted here".
So the offset between the bluest colour and the innermost contour **measures how far the prior pulls the answer away from where the data alone would put it**. Here that offset always points the same way: the likelihood wants shorter residence times and larger inputs than we report.
Three honest limits. (i) The colours are prior-free in *value* but not in *coverage*: each cell shows the best of the draws the sampler happened to visit, and the sampler was steered by the priors. It is therefore a lower bound on the true best fit, and it is weakest exactly where draws are thin — the band around the published value holds 121 141 draws for Yasso20 but only 741 for Yasso15 and *none at all* for Yasso07. A blank cell means *not visited*, never *cannot be fitted*, and Yasso07's blank is precisely why a profile likelihood is needed. (ii) The colour surface is estimated on a coarse grid — one whose cells hold enough draws to support a maximum — and then bilinearly interpolated for display, so it looks continuous without claiming resolution the draws cannot support. (iii) Colour is the log-likelihood *deficit* from the best fit found, clamped at 25 units: a deficit past that is decisively rejected, so resolving 30 from 200 would convey nothing and would flatten the few units of structure along the ridge that actually matter.

To be redone before submission. Replace the sampled colour map with a true *profile likelihood* — an optimisation over the remaining parameters at each grid node — which removes the prior from the coverage as well as from the value, prices the region beyond the posterior's reach, and is smooth enough to interpolate. Roughly 7 h per model on a 20×20 grid; **Yasso07 first**, since it is the model for which no sampled estimate exists at all. Expect the figure to change again once the σ_{input} prior is tightened.

4.6 Why the climate response owns Yasso07's gap and cannot own the others'

The lower row of Fig. 11 shows that residence time is made uncertain by different things in different models: climate dominates in Yasso07, transfer fractions in Yasso15, and the two share Yasso20. That asymmetry looks at first like an artefact of the priors. It is structural.

Yasso07 applies a *single* climate modifier ξ to every pool, so ξ is a pure rescaling of time and $\text{MRT} = \text{MRT}_{\text{ref}}/\xi$ exactly. Calibration moves ξ from 0.855 to 1.822 ($\times 2.13$) at the Finnish mean climate, and that alone predicts a residence time of 15.7 yr against an actual 15.2 — the entire gap, with nothing left over. The model has nowhere else to put a climate misfit. Yasso15 and Yasso20 carry *three* pool-specific modifiers ($\xi_{\text{AWE}}, \xi_{\text{N}}, \xi_{\text{H}}$); residence time is governed by the slow humus pool, which has its own β_{H1}, and that modifier moved only $\times 1.04$ and $\times 1.06$. In Yasso20 the components even move in opposite directions ($\xi_{\text{AWE}} \times 0.76$ against $\xi_{\text{N}} \times 1.16$) and largely cancel. The leverage Yasso07 hands to climate is structurally unavailable to its successors.

Two consequences. First, Yasso07's *published* ξ is $0.855 < 1$: its published parameterisation says Finnish conditions decompose *more slowly* than its reference, where Yasso15 and Yasso20 already say faster (1.16–1.66). Yasso07 starts furthest from what the Finnish data want and is the only variant able to travel the whole distance in one parameter. Second, calibration **inverts the family ordering**: published 33.5/30.4/19.0 yr (Yasso07 slowest) becomes 15.2/22.1/17.5 (Yasso07 fastest). Reproduce with `doublechecks/xi_published_vs_ours.R`.

4.7 The frontier: where the recoverable signal still hides

If not in the kinetics, where is the signal that remains? A random-forest analysis of the residuals answers the same way across every model: the leading, diffuse, shared driver is stand basal area — a productivity and input proxy, not a kinetic or pool variable (Fig. 12). What is left to explain lives in the forcing, not the kinetics, pointing in the same direction as everything above: the way forward is better inputs, not a bigger model.

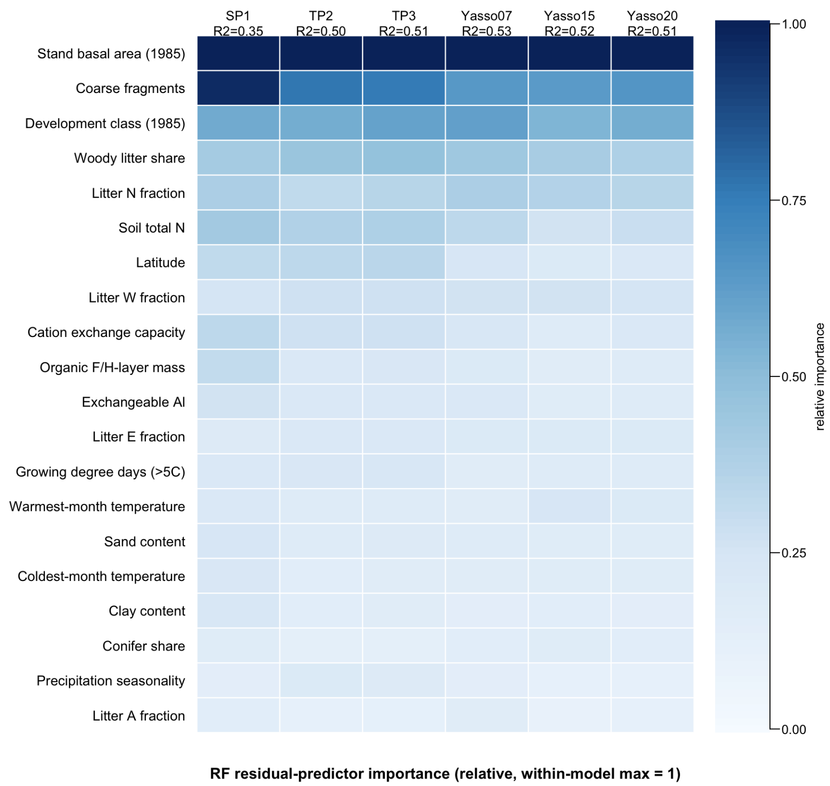


Figure 12: Residual structure. Random-forest importance of candidate drivers for the plot-level residuals (relative, within model). Stand basal area leads in every model; the residual points to site productivity and inputs, not to kinetics.

5 Landing: the policy deliverable

The application is a wall-to-wall national SOC product. The ensemble mean gives a country-wide stock estimate, and its between-model spread maps where the six structures agree and where they do not (Fig. 13) — the spatial face of the forecast divergence, and the reason the uncertainty panel is not an afterthought but the point. Alongside the current rate of change, the pool composition gives a second, distinct axis: how the carbon is held, from fast-cycling, less-processed material to stabilised humus (Fig. 14). What the study hands to the inventory is therefore not a single number but a calibrated estimate together with its irreducible structural uncertainty.

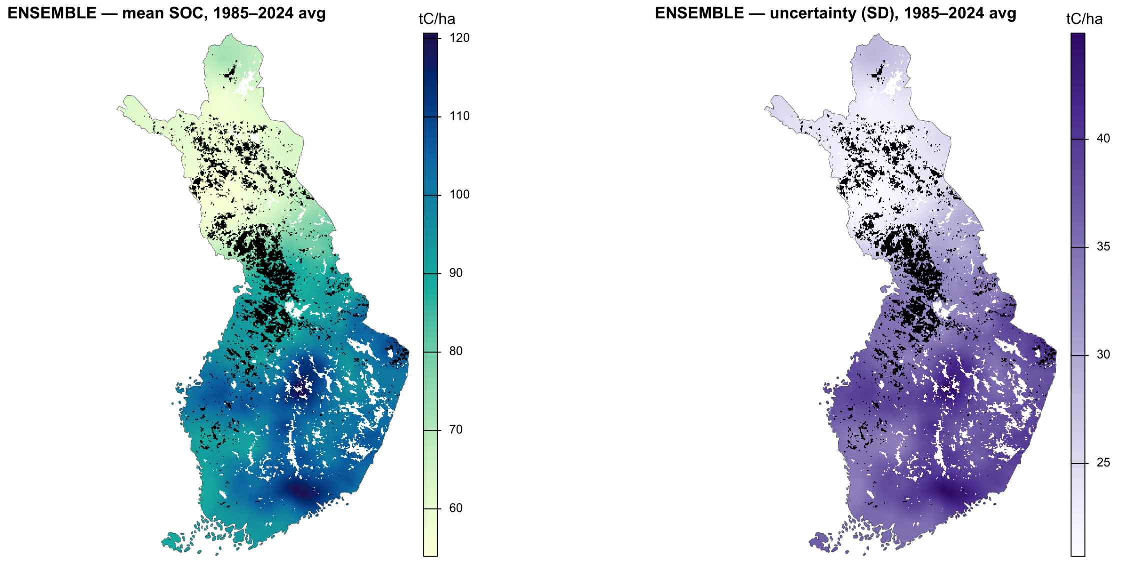


Figure 13: National SOC and its honest uncertainty. (a) Kriged ensemble-mean SOC over Finland; (b) the between-model spread per cell — modest but structured, the spatial expression of "structure is a weak lever."

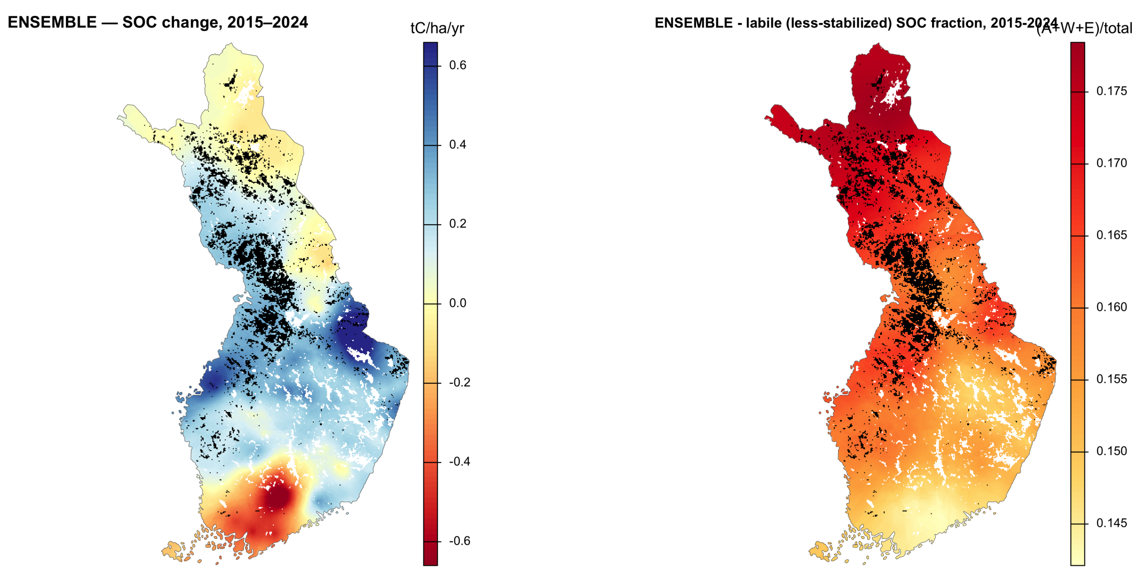


Figure 14: Two axes of soil-carbon dynamics. (a) The 10-year SOC change — gaining vs. losing soils; (b) the labile (less-stabilised) SOC fraction — how the carbon is held. The two are only partly correlated, so (b) adds a distinct signal rather than restating (a).

6 Outlook: closing the loop

The story ends where it began. Solving the initialization problem is what let us quantify the saturating sink; the residual disagreement about *how much more* it will saturate is the uncertainty we hand to policy; and the way to narrow that uncertainty is not more model structure but better forcing — reconstructing the historical input trajectory (replacing our linear guess at the pre-observation phase) and, as an option, stratifying input level by site class. Non-equilibrium is the frame that ties the levels together: it is the cause (a managed forest history), the obstacle (an initial state we cannot assume), and the stakes (a sink still in motion) — one thing, seen from the model side and the policy side at once.

A concrete next step (to explore). The spin-up currently ramps litter *linearly* from the 1917 anchor to the 1985 start. But the national growing-stock history (Fig. 1) already gives the *shape* of that recovery — a depleted mid-century base, then a rise — and litter scales with standing biomass. Using that curve as the interpolation function, in place of the straight line, would turn the history thread from a narrative justification into an actual model input; it is the cheapest realisation of the "reconstruct the input history" frontier, and it targets directly the coarse pre-1985 phase behind the +26 $\mathrm{tC\,ha^{-1}}$ 1985 over-prediction (Fig. 5).

A Supporting figures

These are the supplement: robustness and companion figures that back the main claims without carrying the narrative. They are kept in full so no detail is lost relative to the working storyboard.

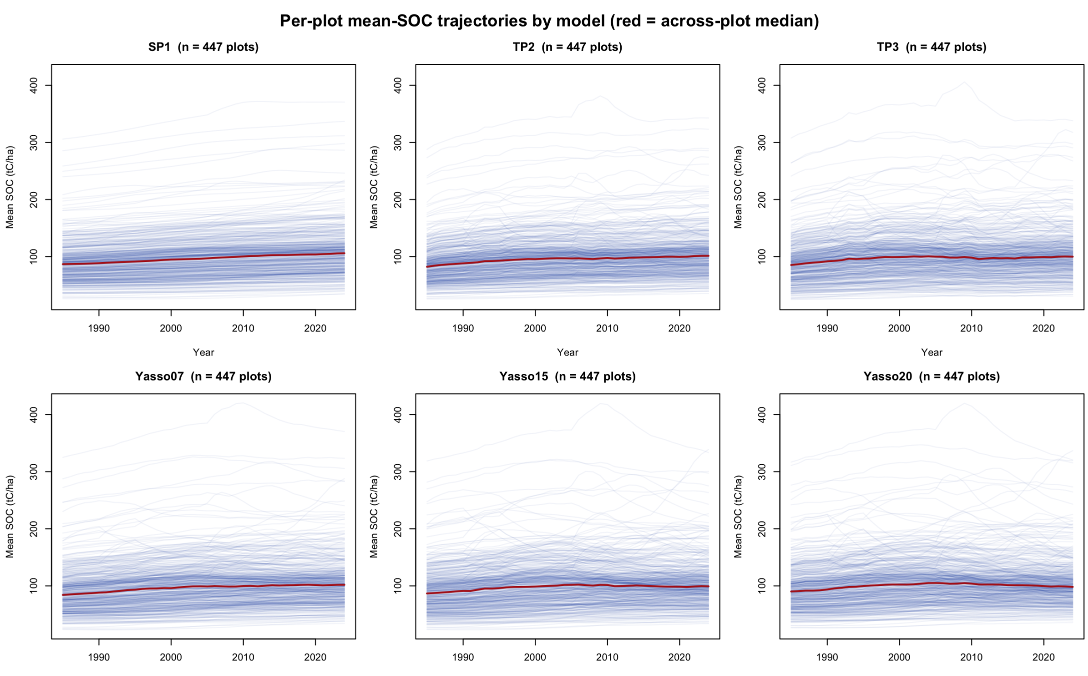


Figure 15: S1 — per-plot SOC trajectories. Every plot's mean SOC by model; all now peak at a physical ~100 $\mathrm{tC\,ha^{-1}}$ (TP2/TP3 jagged, SP1 smooth). Confirms the flux bound produced no inflated stocks and that the TP3 oscillation is a real fast-pool signal, not a numerical artefact.

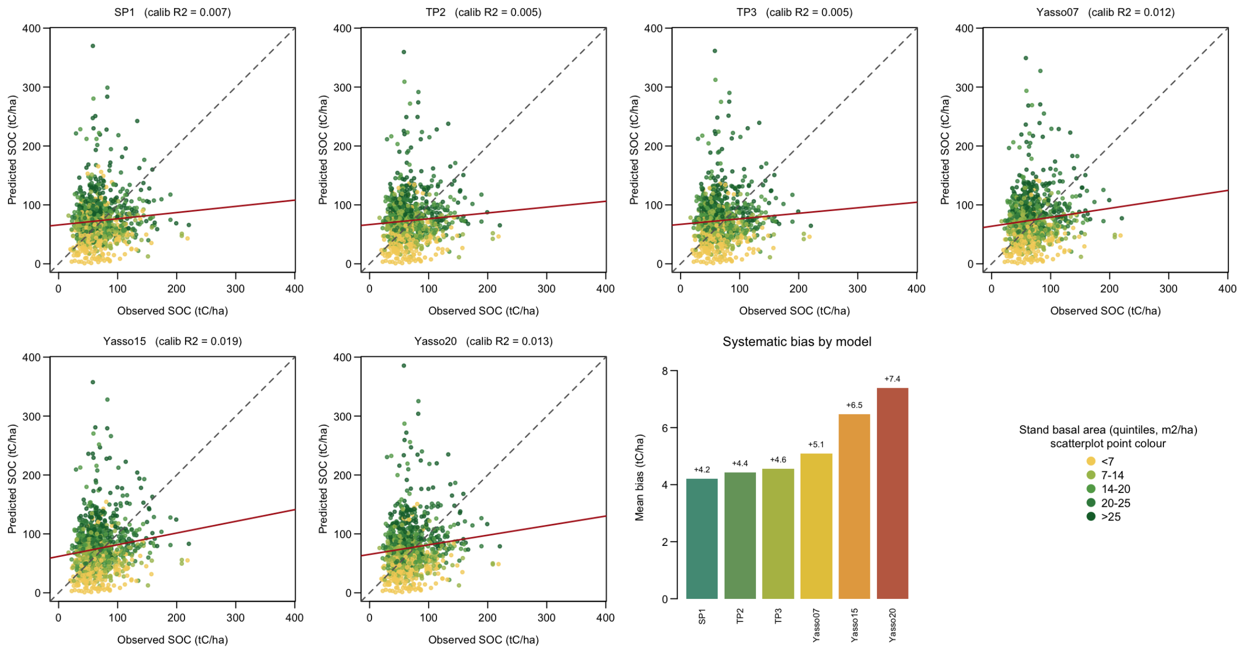


Figure 16: S2 — observed vs. predicted, calibration plots. The in-sample companion to Fig. 6 (six scatterplots coloured by basal-area quintile, as there) plus a systematic-bias barplot in the per-model palette. Same small spread, one notch higher than holdout; bias highest for Yasso20.

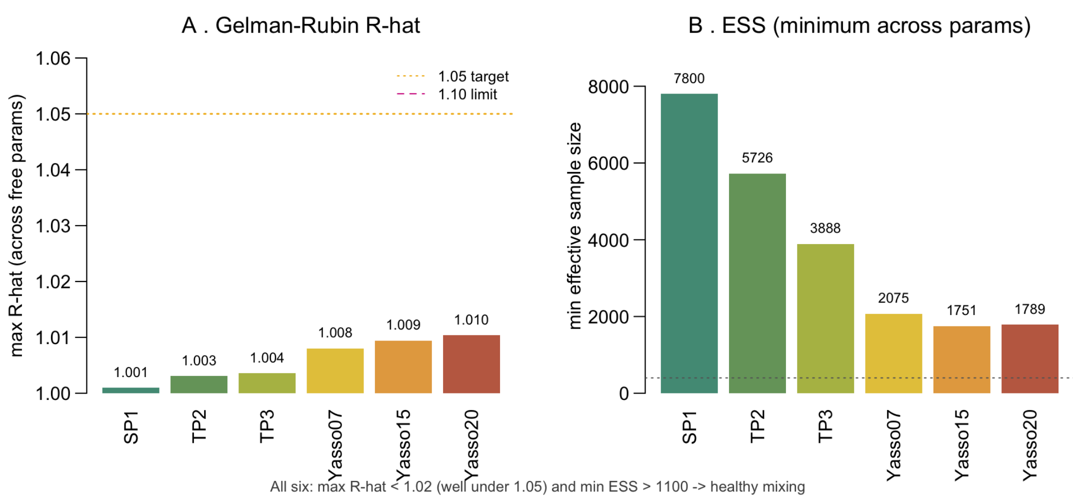


Figure 17: S3 — MCMC convergence. Per-model maximum R-hat (<1.02 everywhere, well under 1.05) and minimum effective sample size (>1100). Good mixing — so well-behaved fraction R-hat reflects mixing, not identifiability (the non-identifiability shows up in information gain and prior-pinning, not in R-hat).

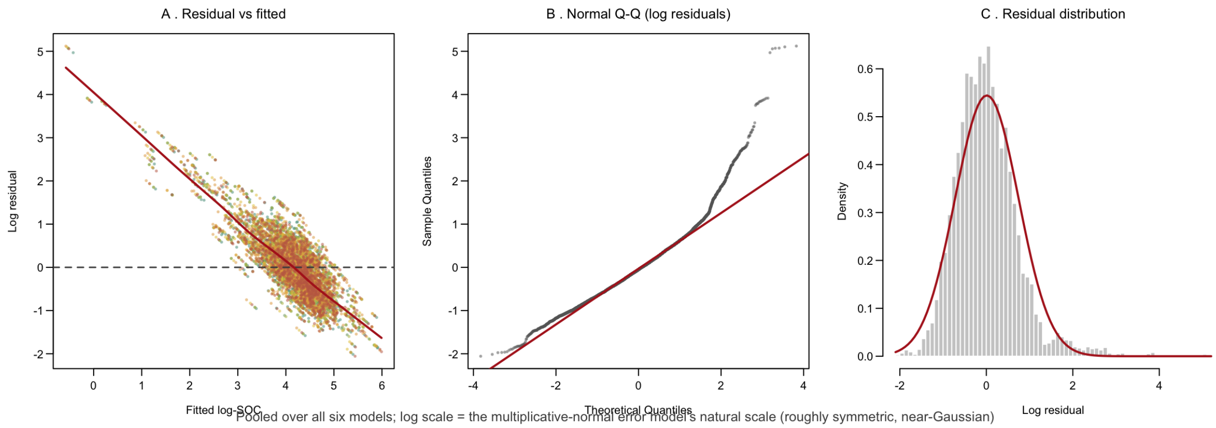


Figure 18: S4 — residual diagnostics. Residual-vs-fitted, normal Q–Q, and histogram of the log residuals, pooled over all six models (log is the multiplicative-normal error model's natural scale). Roughly symmetric and near-Gaussian; the small negative offset (-0.115) is the systematic $\sim$11% over-prediction — a likelihood property, not biogeochemistry.

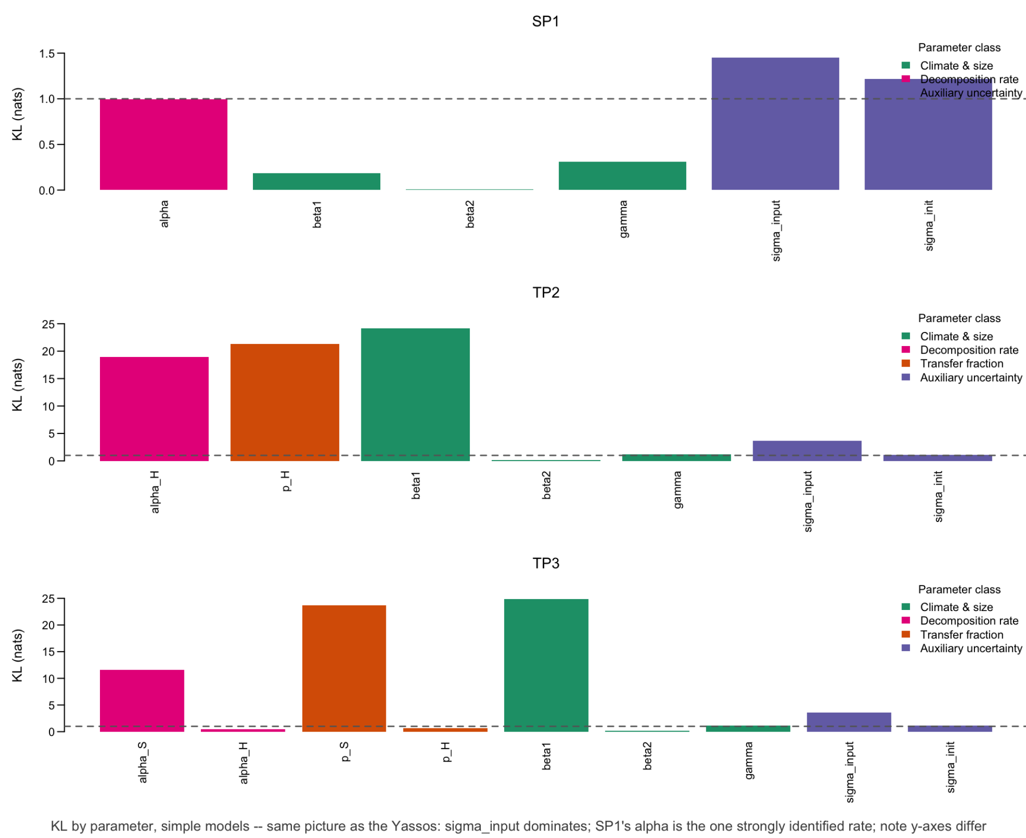


Figure 19: S5 — information gain, the simple models. KL(posterior $\|$ prior) for SP1/TP2/TP3 — the companion to Fig. 9. Same verdict: σ_{input} dominates; SP1's single decomposition rate is the one strongly identified parameter in the study; TP2/TP3 rates gain $\approx$0.

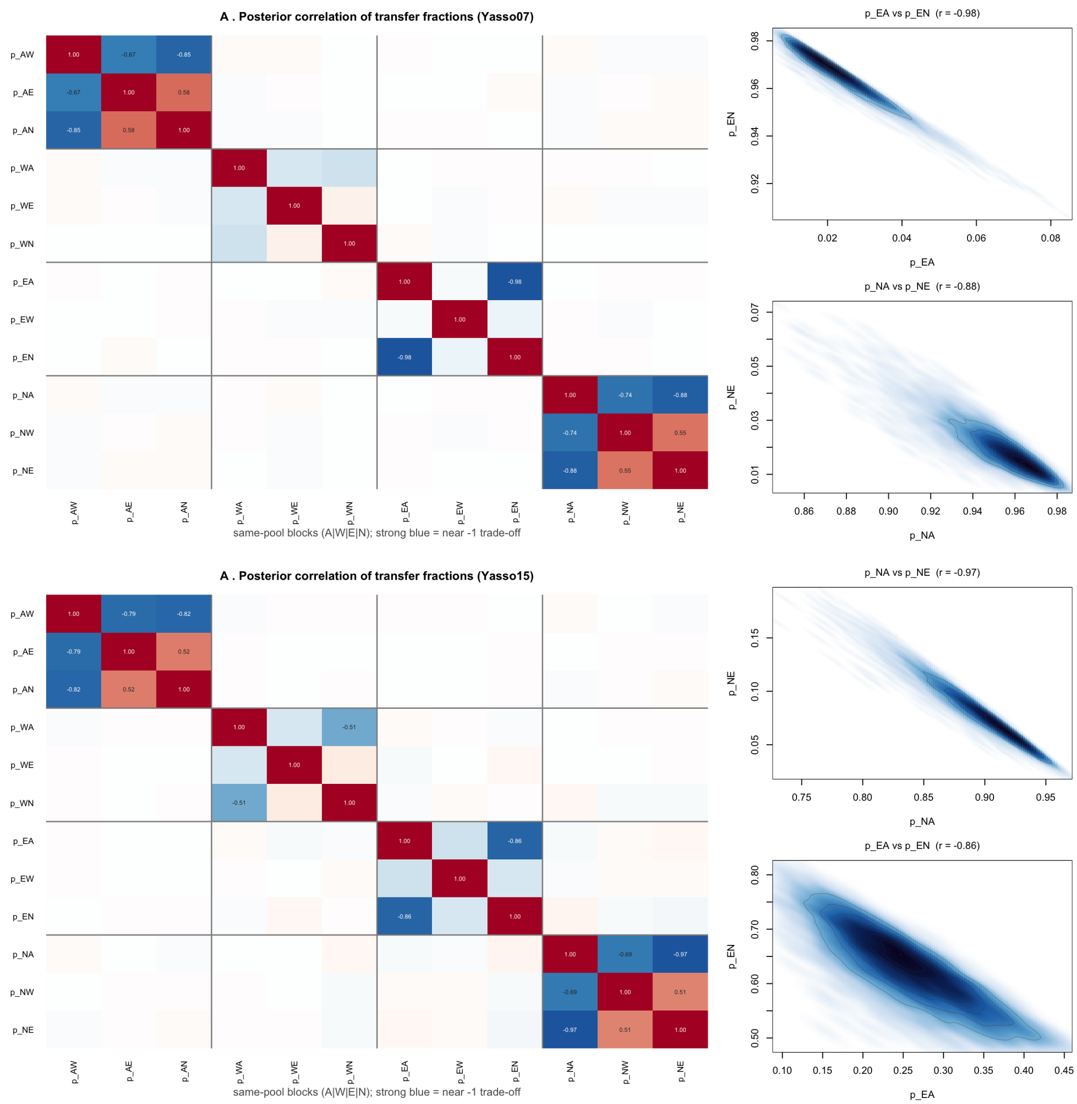


Figure 20: S6 — fraction-ridge structure, the other two Yassos. The same correlation heatmap and density ridges as Fig. 10, for Yasso07 (top) and Yasso15 (bottom). Identical block structure — so the non-identifiability is a property of the model class, not a single variant.

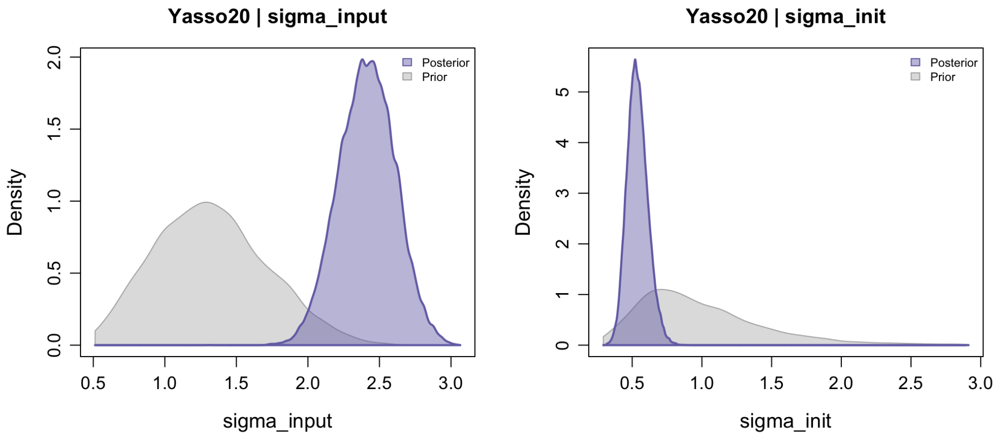


Figure 21: S7 — marginals for the identified auxiliary parameters. Posterior (purple) vs. prior (grey) for σ_{input} and σ_{init} (Yasso20). Both move decisively off the prior — the counterpoint to the non-identified transfer fractions: where the data *do* constrain a parameter, it shows. (More parameter marginals can be added.)

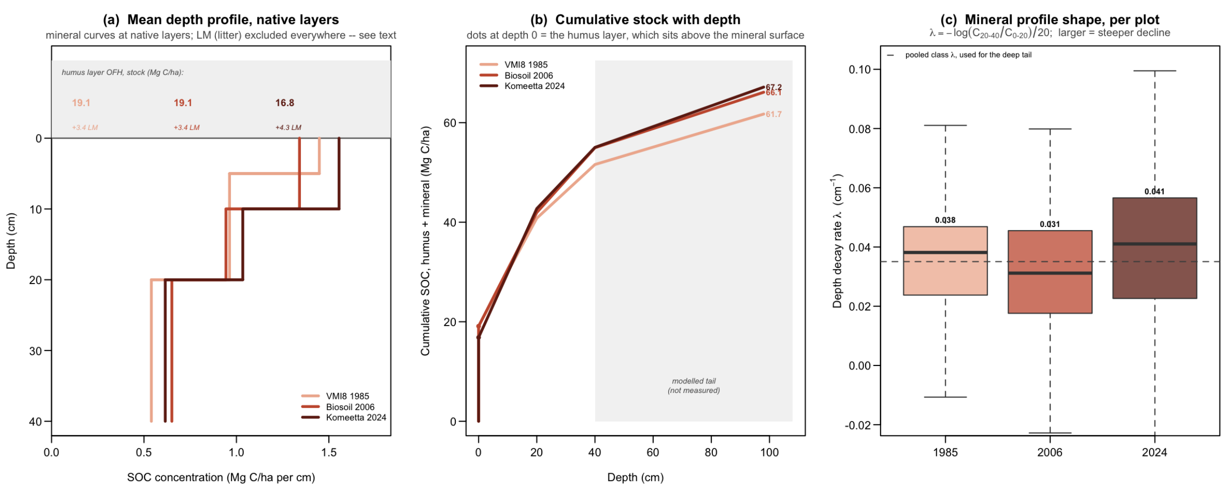


Figure 22: S8 — SOC depth distribution of the three campaigns. All panels on one basis: humus (OFH) + mineral, with the litter layer (LM) excluded because it is absent from the 1985 record; its size is printed in (a). (a) mean profile at each campaign's *native* layers (VMI8 splits 0–20 as 0–5/5–20, the later campaigns as 0–10/10–20). (b) cumulative stock with depth, the sub-40 cm tail shaded as modelled, not measured. (c) the interval-free shape statistic $\lambda = -\log(C_{20\text{-}40}/C_{0\text{-}20})/20$, medians 0.038 / 0.031 / 0.041 cm^{-1}. **1985 sits between the other two**, so the profile *shape* does not single it out — the campaign difference is a level, not a distortion of the depth distribution.

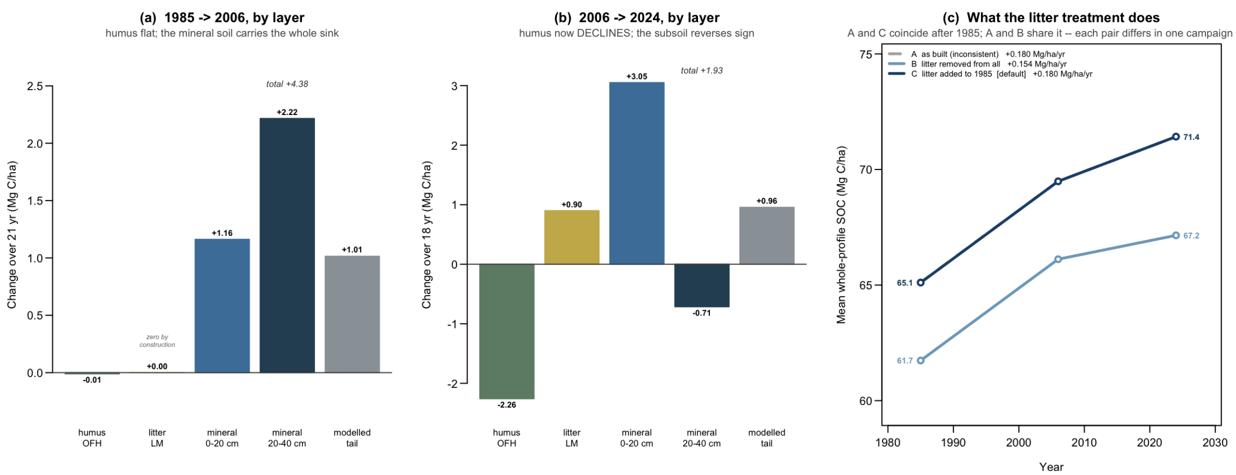


Figure 23: S9 — where the apparent SOC change sits, by layer. (a,b) change per layer over each inter-campaign interval. **The humus layer is flat over 1985–2006 (−0.01) and *declines* over 2006–2024 (−2.26), while the mineral soil carries the entire apparent sink and the subsoil reverses sign** (+2.22 then −0.71). A genuine accumulation signal should be surface-weighted; this one is depth-weighted. (c) the litter treatment as a bracket on the observed trend: A as built (1985 without LM, 2006/24 with — internally inconsistent), B litter removed from all three, C litter added to 1985 (working default). Panels (a,b) are on basis C, under which the 1985–2006 LM bar is zero *by construction* and is drawn hollow to say so. Conditional on the 1985 record being OFH-only (see text).

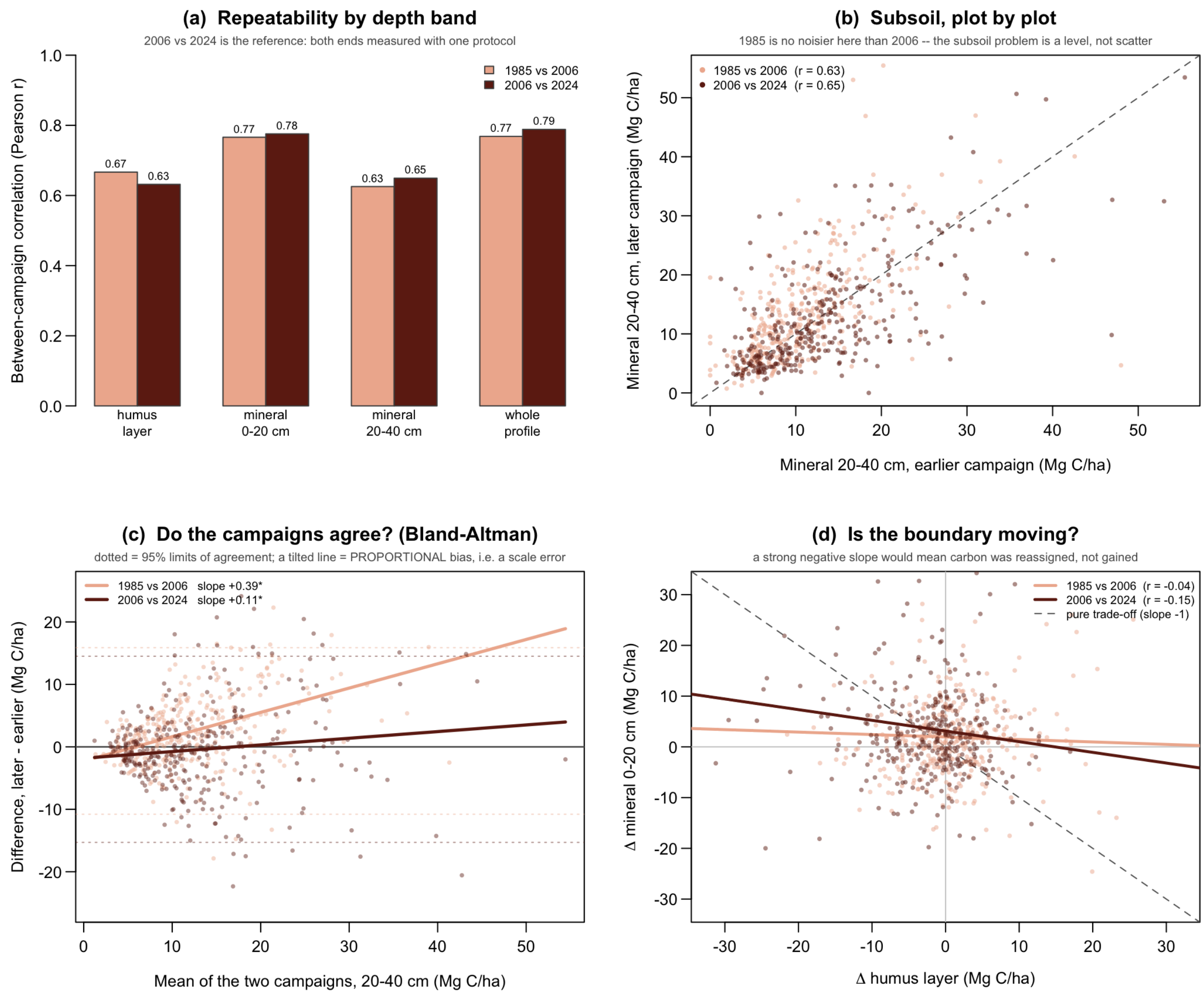


Figure 24: S10 — campaign repeatability, agreement, and the organic/mineral boundary. (a) between-campaign correlation by layer; 2006 vs 2024 is the reference pair, both ends measured under one protocol. (b) the subsoil plot by plot: **1985 is no noisier than 2006** ($r = 0.63$ vs 0.65), so the subsoil problem is a level offset, not scatter. (c) Bland–Altman agreement for the subsoil — the difference between campaigns against their average (neither is a gold standard). A non-zero intercept is a constant bias; a *tilted* line is proportional bias, the fingerprint of a scale error (bulk density, sampled volume) rather than an added amount of carbon. 1985–2006 slope +0.39, equivalent to $C_{2006} \approx 1.48\,C_{1985} - 2.8$. **Note the limits of agreement, -10.8 to $+15.9$ Mg C ha^{-1} on a layer holding $\sim$11: the mineral data cannot resolve plot-level change in either pair.** Proportional bias is present in *every* campaign pair and both mineral layers (0–20 cm: +0.17 for 1985–2006, +0.34 for 2006–2024), so it is a property of the mineral measurement in general, not of 1985 alone. (d) the boundary test — the organic/mineral split is set by eye in the field, so carbon can be reassigned between the humus and mineral 0–20 cm bands with no carbon moving; a pure reassignment would follow the dashed slope -1. Observed $r = -0.04$ (1985–2006) and -0.15 (2006–2024): a weak effect in the later pair only.